

Lose the Faces, Keep the Lesson: Expression-Preserving Face De-Identification for Educational Research

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Abstract. *We propose a practical framework for processing classroom lecture recordings that de-identifies students' and instructors' faces by replacing them with synthetic, demographically similar ones while preserving expressions, gaze and affect. The motivation is to unlock large-scale, ethically usable audiovisual data for teacher training and educational research, capturing authentic classroom dynamics such as communication patterns, interaction, and teaching styles without exposing personal facial information. Building on recent advances in convolutional and generative modeling, we combine high-recall face localization with an identity-removal pipeline focused on retaining emotional cues and structural coherence across frames. Our evaluation plan considers both privacy metrics (re-identification risk and identity similarity reduction) and utility metrics (expression fidelity and pedagogical analyzability). Expected outcomes include a unified framework, offering privacy-preserving yet realistic footage suitable for analysis, annotation and observation of real world educational practice.*

1. Introduction

In an era of super-information, where attention is scarce and students' cognitive loads are constantly taxed, teaching methods must adapt, making research on real classroom dynamics essential. High-value evidence lives in lecture recordings: authentic interactions, affection, turn-taking, and instructional movement that cannot be reconstructed in synthetic settings. Yet the same footage that enables rigorous analysis exposes the identities of children, teenagers, and teachers, creating ethical and legal barriers for usage in large-scale studies and sharing. The central problem, therefore, is to preserve the behavioral signal (expressions, reactions, subtle socio-emotional cues) while removing personally identifying facial attributes so that the material can be safely disseminated and analyzed.

This project advances a practical answer. We posit the following hypothesis: replacing faces in classroom videos with synthetic, demographically consistent faces generated by diffusion models (conditioned on the detected expression and pose) will (a) reduce identity similarity below a strict threshold (low re-identification risk) while (b) preserving emotion/expression cues within acceptable drift for downstream pedagogical analysis. In other words, we explicitly relate the proposed method (expression-aware, diffusion-based face replacement following high-recall detection) to two measurable outcomes: privacy protection and analytical utility. The study will gather evidence for or against this hypothesis; the written work will present arguments and results that support or refute it.

Our objective is to design, implement, and validate an end-to-end framework that detects and localizes faces with high recall in lecture settings and then performs expression-preserving de-identification through synthetic face replacement, maintaining

pose and scene coherence. We aim to (i) ensure that no face goes unnoticed under typical classroom conditions (occlusions, board turns, motion), (ii) achieve identity suppression according to embedding-based and attack-based privacy metrics, and (iii) retain affective/behavioral signal according to automated emotion agreement and human-rated judgments of pedagogical analyzability. The intended outcome is an accessible, compute-efficient pipeline that schools and research labs can run on standard hardware, enabling ethically shareable yet analytically faithful classroom video at scale.

The scope is deliberately focused: lecture recordings in classrooms, frontal cameras, with audio preserved and face level de-identification as the primary target. We do not initially address full body anonymization, lip synchrony preservation for fine grained speech analysis, formal privacy guarantees, or large-scale cloud orchestration; these are acknowledged limitations and avenues for follow-on work. Additional risks include extreme occlusion and demographic fairness in synthetic face generation; both will be tracked empirically and discussed with corresponding governance measures (consent, storage policies, auditability). Within these bounds, we hope to contribute a complete, evaluable framework that directly addresses the field’s core trade-off: privacy strong enough to share and signal rich enough to study.

2. Related Works

With face detection evolving from traditional feature engineering to deep learning, we can now apply CNN-based methods such as YOLOFace and RetinaFace, both analyzed by [G. F. Ananda and Ardiyanto 2024] in a comparison well suited to our classroom environments, showing how the two networks excelled in precision and recall (near-perfect scores). The YOLO family performed better in inference efficiency, illustrating how a single network over the entire image yields impressive speed–accuracy results, meanwhile, RetinaFace, by leveraging deep residual backbones with integrated pixel-level face localization, achieved state-of-the-art accuracy and recall with strong resilience to challenging conditions (diverse lighting and occlusion), but exhibited the worst inference time among the compared networks. Importantly, while many detectors are validated on standard benchmarks (LFW [Huang et al. 2008], FDDB [Jain and Learned-Miller 2010], WIDER FACE [Yang et al. 2016]), [G. F. Ananda and Ardiyanto 2024] bridge this gap by testing on a collected classroom dataset, capturing fluctuating lighting and diverse student postures often absent from those benchmarks.

The results [Ahmed 2025] comparing YOLOv9 to YOLOv5 and YOLOv8 show clear gains in precision, recall, and overall detection quality, especially in classroom-like conditions, making YOLOv9 (notably the medium model) a strong fit for our detect then anonymize pipeline. For our non real time use case, accuracy matters more than marginal latency, so YOLOv9’s higher fidelity supports reliable face localization without sacrificing expression fidelity downstream. Their study trained and evaluated on a custom, manually labeled dataset of children’s faces collected from public websites.

On the other hand, face anonymization does not offer a single recipe. High-fidelity image synthesis remains challenging and newer generative approaches broaden the design space but also complicate evaluation. Earlier work largely relied on encoder–decoder families and k-anonymity-inspired de-identification (e.g., k-Same variants), trading identity suppression against visual utility. More recent systems aim for

selective privacy—removing identity while retaining attributes such as expression and pose, yet faces open questions on guarantees.

As the literature moved beyond k-Same toward generative pipelines, [Wu et al. 2019] set the tone by making the privacy–utility balance explicit: a conditional GAN is steered by a verifier (contrastive loss on face embeddings) to push outputs away from the source identity, and a regulator (SSIM) to keep luminance/contrast/structure, so privacy and usability are optimized together rather than left to adversarial training alone. Building for higher fidelity and control, G2Face fuses a strong generative prior with an explicit 3D geometric prior and an identity-aware fusion module, swapping identity while preserving pose, landmarks, and expression; it even offers a reversible “password” path, though at higher complexity and without video modeling [Yang 2024].

With an expression-first stance, GANonymization compresses the face to dense landmarks and re-synthesizes from that minimal carrier, frequently preserving emotion better than generic GAN baselines while meeting attack-style anonymization thresholds; the trade is reliance on landmark sufficiency and image-only scope [Hellmann 2024]. Its evaluation spans AffectNet [Mollahosseini et al. 2019] (0.4M in-the-wild images across eight emotions), CK+ [Lucey et al. 2010] (593 sequences at 30 fps) and FACES [Ebner et al. 2010] (2,052 high-quality frontal images). Each dataset is paired with anonymized counterparts produced by GANonymization, DeepPrivacy2 [Hukkelås and Lindseth 2023], and CIAGAN [Maximov et al. 2020]. Read in sequence, these works sketch a clear evolution: from explicitly modeling the trade-off (PP-GAN), to geometry-aware identity control (G2Face), to task-oriented carriers for affect (GANonymization).

What GANs buy us is photorealism with controllable identity suppression and a natural place to encode the privacy–utility trade-off and measure it with attack-based privacy and utility on attributes/affect; what they don’t yet deliver are formal guarantees, temporal consistency, and full protection from dataset and demographic biases, and they can be computationally heavy or brittle under occlusion and rare poses.

Diffusion models present a promising alternative to GAN-based approaches in facial anonymization. One notable method [Kung 2025] uses a latent-diffusion U-Net with “ReferenceNet” branches so the synthesized face remains coherent (pose, gaze, expressions, background) while a single anonymization knob controls distance from the source identity. Practically, ReferenceNet mirrors the U-Net to inherit Stable Diffusion v2.1 weights, and the model is trained on CelebRef-HQ [Zhu et al. 2022], CelebA-HQ [Karras et al. 2017], and FFHQ [Karras et al. 2020] with triplets (source, ground-truth, and a face-swapped “driving” image) to learn controlled identity displacement. Training fine-tunes the U-Net and only the attention blocks in ReferenceNet. The result is strong identity suppression with high attribute fidelity, but it does not always beat StyleGAN-inversion baselines on re-ID, shows drops for underrepresented groups, and omits temporal modeling. As an image level baseline it fits classroom needs for privacy control and expression retention, while leaving video consistency and fairness to future work.

In contrast, another diffusion based approach focuses on active identity manipulation through gradient injection during the reverse diffusion process [Wang 2025]. This

method inverts a source face into the latent space and uses gradient-based guidance to gradually move the features toward a target identity while promoting divergence from the source. U-Net’s self-attention layers preserve facial structure, and late-stage truncation enhances the visual quality of the generated face. The result is a highly transferable adversarial face, effective at impersonation. Verification is tested on CelebA-HQ [Karras et al. 2017] and LADN [Gu et al. 2019] using grouped source–target pairs, and identification on 500 CelebA-HQ probe–gallery pairs, targeting standard FR backbones under black-box conditions and probing two commercial APIs. The focus remains targeted impersonation rather than expression-preserving de-ID, with no temporal modeling or formal privacy or fairness analysis, so adapting it to video would require additional consistency constraints.

3. Methodology

This is a theoretical–practical (empirical) study with a primarily quantitative approach. The pipeline normalizes input video, performs high-recall face detection and tracking tuned to classroom-like conditions, estimates per-frame affect/expression signals, and applies a generative de-identification step that replaces each detected face with a non-identifying surrogate while preserving pose, expression, and scene coherence. Temporal consistency and basic post-processing ensure stable outputs suitable for downstream analysis.

Evaluation is dual-objective. On the privacy axis, we quantify identity suppression via embedding-space similarity before and after de-identification, and run attacker-style tests: (i) verification ASR = fraction of de-identified faces still verified as the target, (ii) identification Rank-N-T = target appears in top-N of a closed-set gallery, and (iii) API confidence from commercial FR services (all per [Wang 2025]). On the utility axis, we test whether behavioral signal is preserved: categorical emotion agreement and continuous distances for face shape/expression (L2 over 3DMM coefficients), pose/gaze (quaternion angular distances), plus image quality (PSNR, SSIM, FID; and a face-specific IQA) following prior anonymization benchmarks [Wang 2025] [Kung 2025]. Reproducibility and ethics are addressed through documented configurations, synthetic exemplars for illustration, and explicit clarification.

4. Chronogram

It is clear the face recognition/localization piece is the most mature, so Phase 1 focuses there: review and select datasets (also assess collecting UFPR classroom video with proper consent), select a detector/FR stack, then test, fine-tune, and report metrics on the chosen sets. We’ll then standardize the output per frame, return face crops and bounding boxes with IDs/confidence and expose a simple input API for the generative stage. Phase 2 tackles anonymization: review GAN/diffusion options, choose two candidates, and integrate them with frame-to-frame controls (reuse previous synthetic face, optional gradient-injection for consistency) into the pipeline. After selecting one primary method, we lock the evaluation plan and metrics (privacy and utility) and run the full study, then consolidate results and write-up. We don’t aim to invent a new generator, the goal is to make existing models video-capable and documented—leaving model improvements for future work.

Table 1. Chronogram: (total $\approx$ 20 weeks)

Phase	Task / Goal	Expected Output	Window
1	Review and select datasets; assess UFPR classroom collection with consent	Final dataset list	Weeks 1–2
1	Select detector/FR stack	Baseline FR pipeline	Week 3
1	Test, fine-tune, and report sets metrics	ID metrics report	Weeks 4–5
1	Standardize outputs and API	Per-frame crops, boxes; input API for gen stage	Week 6
1	Deliver censorship baseline (blur/mosaic)	Ready-to-use censored video pipeline	Week 7
2	Review GAN/diffusion options; pick two candidates	Shortlist of 2 generators	Weeks 8–9
2	Integrate candidates with frame-to-frame controls	Working replacements; option to reuse previous face	Weeks 10–13
2	Down-select primary method; lock evaluation plan	Chosen generator; fixed privacy/utility metrics	Week 14
2	Run full study on test set	Results	Weeks 15–16
2	Consolidate results and write-up	Final analysis and documentation	Weeks 17–19
2	Reproducibility and ethics pass	Configs, seeds, scripts, synthetic exemplars	Week 20

5. Expected Results

Technically, we expect a unified and accessible framework that delivers high-recall detection and expression-preserving de-identification suitable for classroom-style video, achieving substantial identity-similarity reduction and low targeted re-identification relative to blur/mosaic, while keeping emotion/expression agreement within a small, predefined loss margin. Longer-term consequences include easier inclusion of authentic video evidence in teacher development and a solid base for extensions such as broader body-cue handling, tighter guarantees, and scaled deployment once data-governance resources are available.

Frame-to-frame detection (e.g. YOLO, RetinaFace) to localize (and optionally track) every face. The output is a per-frame list of identities/track IDs and their bounding boxes with confidence scores, plus a rendered “censorship” version (blur/mosaic/black box) that can already be used ethically. Priority is high recall and stable boxes in class-

room conditions; real time is not required.

Once the faces are localized, a diffusion or GAN based model will be applied to generate new, anonymized faces that preserve expressions and other key features, then replace the original faces in each frame. To address temporal consistency, the previously generated face could be retained and used as a reference for the next frame, combined with the gradient injection technique [Wang 2025] to guide the identity towards the target while maintaining facial structure. However, ensuring the visual continuity of facial features over time remains a critical challenge, requiring further refinement in the integration of the face generation and gradient-based manipulation processes.

We also make explicit what is out of scope so expectations align with deliverables. We adopt image-level, expression-preserving face replacement with temporal smoothing rather than a fully video native generative model and true sequence modeling (frame-to-frame) or identity locking remains future work. We do not claim formal privacy guarantees (e.g., differential privacy or provable unlinkability). Privacy is evaluated attack-wise via embedding similarity and re-identification risk. Fairness will be reported with stratified analyses by demographics and scene conditions, but cannot be guaranteed across all subgroups given dataset limitations. Non-face identifiers such as voice, clothing logos, body shape, and background context are not anonymized in this release. These boundaries, together with insights from GAN and diffusion-based literature, guide both our design choices and our evaluation protocol.

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